6 Workload-Level Budget Calibration

The policy selects one workload-level refinement multiplier through unlabelled agreement with a larger-budget reference, then reuses it across batches. Quality and calibration cost constrain that reuse.

6.1 Why Fixed Refinement Budgets Can Be Wasteful

For routed universe $U(q)$, JHQ's refinement multiplier sets [16]

$$c(\alpha) = \max\{k, \lceil \alpha k \rceil\}, \qquad C_\alpha(q) = \mathrm{Top}^{\mathrm{primary}}_{\min\{c(\alpha),|U(q)|\}} U(q). \tag{6.1}$$

C_α contains valid survivors; underfull output slots receive sentinels. Analysis requires at least k valid candidates. α changes retention within U, while probes change U itself.

Too much refinement wastes residual reads; too little can lose neighbours. The grid spans 20–2,000 survivors at $k = 10$ without retraining. Smaller budgets do not release stored residuals or necessarily shrink previously grown workspace.

For fixed batch and routing, the cost split is

$$T(\alpha) = T_{\mathrm{base}} + T_{\mathrm{select}}(c(\alpha)) + T_{\mathrm{refine}}(c(\alpha)) + T_{\mathrm{final}}(c(\alpha)). \tag{6.2}$$

T_{base} covers transfers, rotation, routing and primary scores; survivor-dependent work follows figure 6.1. Capacity/resource thresholds prevent a monotone-latency claim. The rule minimises accepted budget, not noisy QPS (section 7.5.5).

Each calibration trial still runs rotation, probing and primary scoring before refinement. Reusing the index avoids rebuilding or retraining; it does not cache every intermediate query result across budget trials. The cost of calibration must therefore count full searches, not merely the difference between two residual-scoring workloads.

The policy selects one budget for an index, routing configuration, k and query workload. It uses neither a trained per-query predictor nor an online choice for each arriving query. Scan-group scheduling remains the separate mechanism of section 5.5; its execution path must still be considered when interpreting budget calibration.

Position among adaptive ANN methods. α serves the candidate-expansion role of Faiss's `k_factor` and cuVS refinement [7, 10]. Alternatives use GT tuning [9], learned per-query stopping [19] or predicted recall [6]. Here unlabelled agreement chooses one reusable workload budget. Scores still use reconstructed vectors, unlike the cuVS source-vector refinement example.

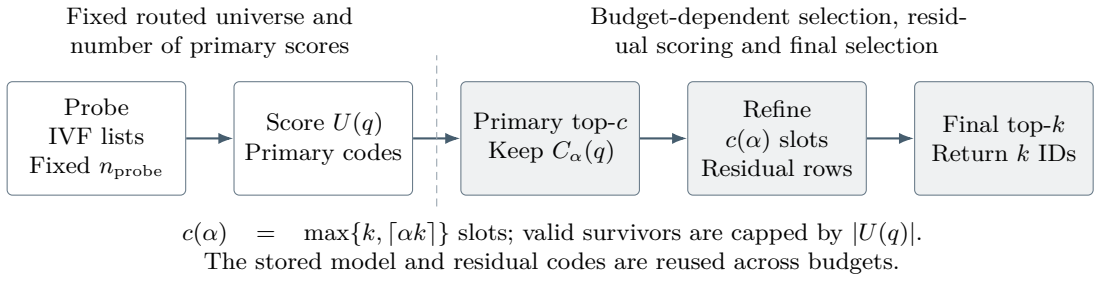


Figure 6.1. The refinement budget changes work after primary-score production. With a fixed query, index and probe count, it leaves the routed candidate universe unchanged. The highlighted stages depend on survivor capacity; this is a logical work decomposition, not a measured latency breakdown. The scan kernel also performs budget-dependent admission and compaction. Padding applies when fewer than c candidates exist.

6.2 GT-Free Sampled Output Agreement

Let $\mathcal{R}_\alpha(q)$ be the set of k valid, unique database IDs returned at budget α on a fixed index. Define $\mathcal{R}_{\text{ref}}(q) = \mathcal{R}_{\alpha_{\text{ref}}}(q)$. The descending grid is

$$A = (200, 100, 64, 32, 16, 8, 4, 2), \qquad \alpha_{\text{ref}} = A[0] = 200. \tag{6.3}$$

For S calibration queries $\mathcal{Q}_S$, define

$$\begin{aligned} D_S(\alpha) &= \sum_{q \in \mathcal{Q}_S} \left(k - |\mathcal{R}_\alpha(q) \cap \mathcal{R}_{\text{ref}}(q)|\right) \\ &= \frac{1}{2} \sum_{q \in \mathcal{Q}_S} |\mathcal{R}_\alpha(q) \triangle \mathcal{R}_{\text{ref}}(q)|. \end{aligned} \tag{6.4}$$

A budget is accepted iff $D_S(\alpha) \le \epsilon$, with $\epsilon = 1$: one missing reference ID across the sample, independent of result order or reported distances. The finite-grid objective is

$$\widehat{\alpha} = \min\{\alpha \in A : D_S(\alpha) \le \epsilon\}. \tag{6.5}$$

The reference supplies a nonempty fallback. The objective minimises an accepted multiplier; section 6.3 states when bisection solves it exactly.

At $k = 10$, the normalised allowance $\epsilon/(Sk)$ is 0.003125, 0.0015625 and 0.00078125 for $S = 32, 64, 128$, and 0.0002 at $S = 500$. Larger S changes coverage and strictness. Disjoint held-out queries assess draws from one finite calibration pool (section 7.5).

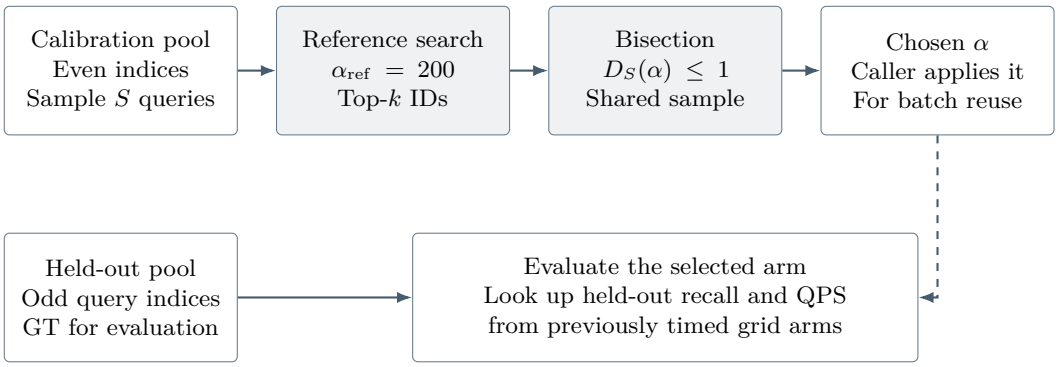


Figure 6.2. Workload-level calibration and its evaluation. Budget decisions use reference IDs; GT assesses held-out quality. Selected-arm results are looked up from timed fixed arms. Implementation boundaries are in section A.3.

6.2.1 What agreement controls

Under Proposition 6.1, disagreement counts reference answers excluded by primary screening:

$$D_S(\alpha) = \sum_{q \in \mathcal{Q}_S} |\mathcal{R}_{\text{ref}}(q) \setminus C_\alpha(q)|. \tag{6.6}$$

For valid k-element outputs, each missing reference ID costs at most one GT hit. On the same sample,

$$\begin{aligned} r_{\mathcal{Q}_S}(\alpha) &\ge r_{\mathcal{Q}_S}(\text{ref}) - \frac{D_S(\alpha)}{Sk}, \\ D_S(\alpha) \le \epsilon \implies r_{\mathcal{Q}_S}(\alpha) &\ge r_{\mathcal{Q}_S}(\text{ref}) - \frac{\epsilon}{Sk}. \end{aligned} \tag{6.7}$$

Scope. This is a sample-local bound relative to the reference. GT evaluation measures the reference's remaining error (section 7.8.2).

6.2.2 Sampling resolution and ordered testing

Let H_S count queries with any missing reference ID. Since $H_S \leq D_S$, acceptance implies $H_S \leq 1$. For a prespecified budget and i.i.d. queries, the one-sided 95% exact binomial upper limit after one event solves [22]

$$(1-u)^S + Su(1-u)^{S-1} = 0.05. \tag{6.8}$$

The query-event limits are 0.03652 at $S = 128$ and 0.009452 at 500. Treating all Sk result indicators as homogeneous independent Bernoulli trials is stronger: the corresponding limits are 0.003701 and 0.0009484, approximately $4.74/(Sk)$. Query i.i.d. alone does not justify dividing the query-event limit by k; neither model certifies repeated subsamples of this fixed pool.

Monotone disagreement makes accepted indices a prefix. Valid per-arm tests can then use fixed-sequence error control [3]. Our one-slot threshold tests stability; it does not certify GT risk.

A finite-pool extension. Assume each query has fixed $X_q(\alpha)$ at the intended execution regime, independent of subset membership/position. For pool size N_{cal} and counts $n_j = |\{q : X_q = j\}|$, uniform sampling without replacement accepts only all-zero samples or one single-count query [23]:

$$\Pr\{D_S(\alpha) \leq 1\} = \frac{\binom{n_0}{S} + n_1\binom{n_0}{S-1}}{\binom{N_{\text{cal}}}{S}}. \tag{6.9}$$

Out-of-range binomial coefficients are zero. Predicting GT risk also needs the grid acceptance structure and held-out fixed-arm quality. Recorded subset sums test count invariance (section 7.5.5).

6.3 Budget Search, Nestedness and Tie Safety

6.3.1 Bisection and complete enumeration

Write $J = |A|$. The implementation computes the sample reference once, then searches for the largest acceptable index in the descending grid:

1. Set $l = 0$, $h = J - 1$. Index zero is the reference fallback.
2. While $l < h$, test $j = \lfloor (l + h + 1)/2 \rfloor$ by running $A[j]$ and computing $D_S(A[j])$.
3. If $D_S(A[j]) \leq \epsilon$, set $l = j$; otherwise set $h = j - 1$.
4. Return $A[l]$. A caller applies it before processing reuse batches.

The eight-point grid needs at most three complete candidate searches in addition to the reference. Invocation, host-comparison complexity and timing boundaries are specified in section A.3.

Enumeration reruns the reference and all seven budgets on the same sample. Both timers include a fresh reference. Enumeration handles acceptance holes; bisection matches its minimum only when accepted indices form a prefix.

Case	C_1	C_2	C_3	Reference overlap
Nested	$\{b\}$	$\{b, c\}$	$\{a, b, c\}$	0, 0, 1
Non-nested	$\{a\}$	$\{b, c\}$	$\{a, b, c\}$	1, 0, 1

Figure 6.3. Constructed $k = 1$ example with refined order $a \prec b \prec c$ and reference $\{a\}$. The nested case preserves reference membership. With all primary scores tied, different tie choices can produce the non-nested middle pool and break monotonicity. These are explanatory sets, not measured GPU outputs.

6.3.2 A sufficient condition for monotonicity

Proposition 6.1 (Nested survivors preserve reference membership)**.**
For one query, assume fixed routing and survivor sets $C_\alpha \subseteq C_\beta \subseteq C_{\alpha_{\text{ref}}}$ whenever $\alpha \leq \beta \leq \alpha_{\text{ref}}$. Assume at least k valid survivors, fixed refined scores and one deterministic total order for final ties. Each result is the exact top-k in that order. Then

$$|\mathcal{R}_\alpha \cap \mathcal{R}_{\text{ref}}| = |C_\alpha \cap \mathcal{R}_{\text{ref}}|, \qquad D_S(\alpha) \geq D_S(\beta). \tag{6.10}$$

The latter statement holds when these conditions hold for every sampled query.

Proof.
For $x \in \mathcal{R}_{\text{ref}} \cap C_\alpha$, the common total order gives

$$|\{y \in C_\alpha : y \prec x\}| \leq |\{y \in C_{\alpha_{\text{ref}}} : y \prec x\}| < k.$$

Hence $x \in \mathcal{R}_\alpha$. Together with $\mathcal{R}_\alpha \subseteq C_\alpha$, this proves the intersection identity in equation (6.10). Nestedness gives $|C_\alpha \cap \mathcal{R}_{\text{ref}}| \leq |C_\beta \cap \mathcal{R}_{\text{ref}}|$. Summing missing reference counts over queries proves $D_S(\alpha) \geq D_S(\beta)$. □

Fixed-key exact primary top-c gives nesting; fixed-key refinement then gives monotone agreement. This supports bisection, without making GT recall or runtime monotone as the pool grows.

6.3.3 A corollary for the evaluated grouped path

Within a draw, S, routing and scan threads are fixed; `scheduled_groups(S)` ignores c. Grouped budgets use the same LUT sums. Fixed B_r, 16 refinement lanes and final $(D_{\text{H}}, \text{ID})$ order preserve each candidate's scoring.

Corollary 6.2 (Grouped calibration)**.**
For the evaluated grouped configuration, under the representation and shared-bound contracts of Chapters 4 and 5, primary survivors are nested across budgets. Consequently Proposition 6.1 applies and bisection returns the same minimum accepted grid budget as enumeration.

Proof.
Lemma 5.1 and propositions 5.1 and 5.2 give exact primary top-c under the common key $(D_{\text{P}}, \text{position})$. Its prefixes nest as c grows. The fixed per-candidate refinement and exact final total order supply the remaining premises of Proposition 6.1. □

The corollary covers 2,176/2,304 draws. The other 128 are OpenAI-3072, $S = 128$, distance-only. All recorded curves are monotone and choices match enumeration (section 7.5.5); that check remains finite.

Full-pool coverage. Two FULL cells group Vogue; ten use $g_{\text{scan}} = 0$. All twelve choices match enumeration. The ten distance-only cells remain outside corollary 6.2. A common cutoff order, or no tie crossing any grid/reference cutoff, suffices for general nesting. Matching held-out execution does not prove that condition.

Table 6.1. Budget controls and the questions they answer. Held-out GT is used to assess every result, but only the diagnostic oracle uses it to choose a budget. The reference is fixed at 200 for agreement-based arms.

Control	Selection information	Question
Fixed 100	No calibration	Gain relative to the inherited default?
Fixed 64 and smaller grid arms	No calibration	Does a simple fixed choice remove the need for calibration?
Sampled bisection	S unlabelled queries; slot agreement	How well does a low-cost sample choose a reusable budget?
Same-sample enumeration	Same S, reference and threshold; all grid arms	Did bisection miss the smallest accepted grid budget?
Full-pool calibration	All 500 calibration queries; same bisection rule	What changes with broader coverage and a stricter normalised allowance?
Held-out oracle	GT recall and timed QPS for all arms	Which measured arm is fastest under the diagnostic quality constraint?

A rank-based extension. With the same nesting/fixed-score conditions and one-based primary rank,

$$D_S(\alpha) = \sum_{q \in \mathcal{Q}_S} \sum_{x \in \mathcal{R}_{\mathrm{ref}}(q)} \mathbf{1}\{\mathrm{rank}_{\mathrm{P}}(x) > c(\alpha)\}. \tag{6.11}$$

Exported reference ranks could recover the whole curve from one search. The current harness instead reruns every budget; rank extraction remains future work.

6.4 Budget Baselines and Their Questions

Table 6.1 separates budget selection from its evaluation. Fixed arms share the index, route and held-out queries; invocation and timing details are in section A.3.

Let $\widehat{r}, \widehat{Q}$ denote held-out recall/QPS and $\widehat{r}^* = \max_{\alpha \in A} \widehat{r}(\alpha)$. The diagnostic oracle is

$$\alpha_{\mathrm{oracle}} \in \arg\max_{\alpha \in A} \left\{ \widehat{Q}(\alpha) : \widehat{r}(\alpha) \geq \widehat{r}^* - \delta \right\}, \qquad \delta = 10^{-3}. \tag{6.12}$$

It uses labels and timings. Its recall slack δ differs from agreement's slot allowance ϵ.

FULL remains label-free bisection and tightens $\epsilon/(Sk)$. Same-sample enumeration checks the search procedure. Agreement minimises an accepted multiplier; the oracle maximises quality-constrained speed, so their choices can differ.

Fixed 64 tests a smaller default; fixed 200 tests a high budget (section 7.5 and table 7.8).

6.5 Hold-out Risk, Calibration Cost and Payback

6.5.1 Choosing a calibration scale

FULL's held-out quality appears in table 7.7; sampled risks appear in figure 7.6. Under the strong independent-slot model, $4.74/(Sk) \approx 10^{-3}$ suggests $S \approx 500$ at $k = 10$. This concerns reference disagreement, not a GT guarantee (section 6.2).

The path and common-order conditions for FULL are addressed in section 6.3.

6.5.2 Risk and selected-arm performance

Over m calibration draws, let draw r select $\widehat{\alpha}_r$. Define held-out loss and threshold-crossing frequency by

$$L_r = \widehat{r}^* - \widehat{r}(\widehat{\alpha}_r), \qquad \widehat{p}_\delta = \frac{1}{m} \sum_{r=1}^{m} \mathbf{1}\{L_r > \delta\}. \tag{6.13}$$

Mean loss, crossing frequency and tail size answer different questions. Table 7.8 and figure 7.6 report the archived summaries; the largest supplementary per-draw losses are reported in section 7.5.5.

Reported selected-arm rate is $\overline{Q} = m^{-1} \sum_r \widehat{Q}(\widehat{\alpha}_r)$, a lookup from timed fixed arms. It is distinct from the median budget's rate and from the throughput of a calibrated query stream. Quality and speed must be read together (table 7.9).

6.5.3 Cost of reuse

For B_{reuse} equal-sized subsequent batches, model

$$T_{\text{policy}}(B_{\text{reuse}}) = T_{\text{cal}} + B_{\text{reuse}} T_{\text{chosen}}, \qquad T_{\text{fixed}}(B_{\text{reuse}}) = B_{\text{reuse}} T_f. \tag{6.14}$$

When $T_{\text{chosen}} < T_f$, the break-even count is

$$B^*_{\text{reuse}} = \frac{T_{\text{cal}}}{T_f - T_{\text{chosen}}}. \tag{6.15}$$

Positive saving requires $T_{\text{chosen}} < T_f$ and $B_{\text{reuse}} > B^*_{\text{reuse}}$. Batch time is $T = 1000n/Q$ milliseconds. When choices vary, $1/\mathbb{E}[Q] \leq \mathbb{E}[1/Q]$, so inverse mean QPS understates mean time. Timing boundaries are in section A.3; section 7.5.6 tests direct calibration-plus-reuse cost on four cells with quality alongside time. Campaign estimates are in table 7.9.

Recalibration every H batches adds T_{cal}/H per batch. Drift detection and automatic recalibration remain future work.

6.6 Policy Failure Modes and Applicability

Reuse assumes a stable index, k, routing and queries. Ordering and batch-path conditions follow section 6.3; quality, tails and total cost follow section 6.5. Drift detection is unimplemented.

The caller must explicitly install α before reuse.

Budget tests and their trace rerun use v69, including OpenAI TF32. FP32 frontier sensitivity does not validate or recalibrate those budget decisions.

7 Experimental Evaluation

The experiments address construction and capacity (RQ1), execution (RQ2), budget quality and reuse cost (RQ3), and competitive search regions (RQ4).

7.1 Experimental Protocol

7.1.1 Comparison protocol and reproducibility

Protocol follows sections 3.2, 3.3 and 3.5. Construction/calibration are separate from host-to-host search; each ablation keeps its own baseline.

CPU JHQ records use unmatched configurations or legacy recall protocols. For example, October Vogue uses 1,024 lists and $[8, 4]$ bits, versus E1's 4,096 lists. They cannot supply a matched CPU/GPU speedup.

Adaptive call counts and auxiliary device-only timing are documented in section A.2.

Metrics and frontier estimators follow sections 3.2 and 3.4; g_{LUT} and g_{scan} retain the distinct meanings in section 5.1.

7.2 Modern GPU ANN Recall–QPS Frontiers

7.2.1 Comparison scope and frontier construction

Figure 7.1 compares JHQ-GPU with the graph, compressed-IVF and brute-force baselines of section 3.4 [17, 28, 33], using the first four datasets in table 3.2. Each uses one batch of 1,000 queries with common GT. T5 separately evaluates small batches in section 7.2.5; section 7.6 reports internal ablations.

The recorded schedule uses $g_{\mathrm{scan}} = 0$ for all 105 arXiv/OpenAI run–budget cells and $g_{\mathrm{scan}} = 4$ for all 27 Vogue cells, triggered by Vogue's candidate-count CV of 0.338–0.622. Thus the non-Vogue frontiers exercise query-centric scanning, while Vogue also exercises grouped retention, shared bounds and merging (section 5.5).

Applying the frontier rule in section 3.4, JHQ contributes 27–39 points per dataset and each baseline family 54–264. E1 contains 1,590 valid points, no failure records and no invalid-ID or duplicate-row flags. These are parameter-sweep points, rather than independent index builds.

The tuning dimensions are listed in table 3.3, with complete coverage in section B.1. A curve ending before a target records missing coverage of that finite grid.

Storage is unmatched: nine-bit RaBitQ has JHQ's nominal $9d/8$ payload; 4/8-bit variants use less. Table 4.1 and section 7.7 give representation costs and measured allocations alongside this performance envelope.

7.2.2 Matched recall and opposing operating regions

Table 7.1 reports recall 0.97 using the interpolated and qualifying-measured estimators defined in section 3.4.

Table 7.1. Recall 0.97 comparison. Each ratio is interpolated / measured; values above one favour JHQ. JHQ kQPS is the interpolated estimate. The small arXiv–RaBitQ difference changes direction with the estimator.

Dataset	JHQ kQPS	vs CAGRA fp32	vs IVF-RaBitQ
Vogue	230.8	1.416 / 1.537	1.059 / 1.046
arXiv	258.6	1.317 / 1.422	0.996 / 1.015
OpenAI-1536	126.7	1.747 / 2.294	1.130 / 1.111
OpenAI-3072	55.9	1.747 / 2.088	1.157 / 1.346

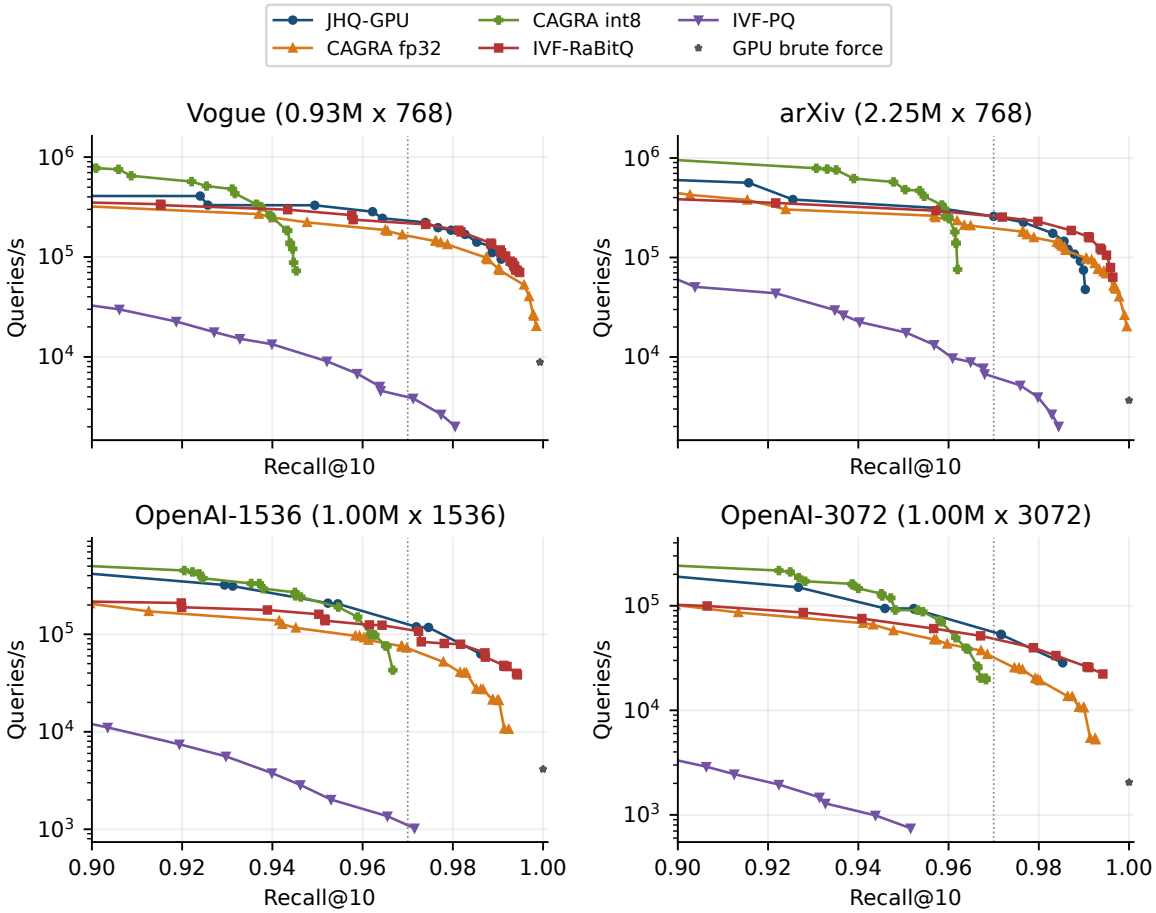


Figure 7.1. Measured host-to-host recall–QPS frontiers on the four E1 datasets. The displayed recall window begins at 0.90, with a vertical line at 0.97; all baseline families, including CAGRA int8, participate in frontier construction. Lines connect measured points; throughput uses a logarithmic axis.

At batch 1,000 and recall 0.97, Vogue's JHQ/RaBitQ ratio falls from 1.059 to 0.933 without grouping. The index, probes, budget and final outputs stay fixed. This identifies an integrated-path benefit, not a shared-bound-only effect.

CAGRA int8 is 1.20–1.98× faster at recall 0.90–0.92, but never reaches 0.97 in these grids. RaBitQ regains some 0.98+ targets; CAGRA fp32 wins near 0.99 on arXiv (figure 7.1).

7.2.3 Precision sensitivity on the OpenAI datasets

Supplementary source inspection establishes that the E1 implementation uses TF32 multiplication with FP32 accumulation for OpenAI rotation and centroid GEMMs, despite FP32 input/storage and a misleading source comment (section 4.2). The main E1 frontier remains the historical measurement of that implementation. In campaign D, a separate FP32 variant changes the cuBLAS math-mode switch and replays 37/39 OpenAI configurations at batch 1,000, five rounds each. Execution order alternates between variants across probe settings. Both reuse the same TF32-trained model; the intervention changes add/query arithmetic, including database encoding and routing. It is not a comparison of independently FP32-trained models. Table 7.2 summarises the paired changes.

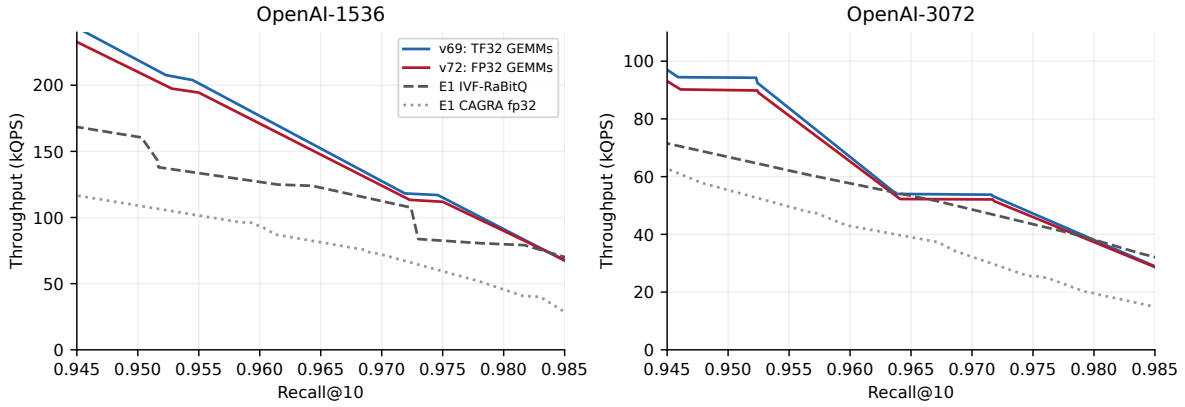


Figure 7.2. Campaign D precision frontiers. Solid curves are same-session JHQ variants; dashed/dotted baselines are historical E1 measurements. All use the chapter's frontier and interpolation rules.

Table 7.2. FP32 sensitivity. QPS ratio is FP32/TF32 at equal configuration (geometric mean); recall differences have the same direction. The last two columns compare the FP32 frontier at recall 0.97 with archived E1 IVF-RaBitQ and CAGRA fp32 frontiers. Campaign D does not remeasure the baselines.

Dataset	Pairs	QPS ratio	ΔRecall	vs RaBitQ	vs CAGRA
OpenAI-1536	37	0.953	+0.0003–+0.0005	1.086	1.679
OpenAI-3072	39	0.951	+0.0000–+0.0003	1.079	1.629

At equal configuration, FP32 reduces throughput by about 5% and increases recall by 0–0.0005. At recall 0.97, FP32 is about 3% slower than same-session TF32, yet 1.086/1.079× archived RaBitQ and 1.679/1.629× archived CAGRA fp32. New-session v69 measurements give 1.118/1.114 against archived E1 RaBitQ, versus historical 1.130/1.157. D's baseline ratios remain cross-session; only v69/v72 are contemporaneous. Figure 7.2 shows the two JHQ frontiers with their archived comparators. A later T1 campaign remeasures baseline anchors in the same session; section B.4.1 reports those separate, grid-sensitive results.

7.2.4 Interpretation and comparison limits

The OpenAI exports are paired widths of Qdrant DBpedia embeddings [31, 32]. After truncation/renormalisation, all 1,000 query rows and 10,299 sampled base rows have cosine at least 0.9980 (median 1.000000); mean GT overlap is 9.046/10. This supports row alignment without original entity IDs. Doubling width keeps the CAGRA ratio at 1.747 and moves RaBitQ's from 1.130 to 1.157. Within this pair, dimension alone does not explain the gain.

Larger gains accompany smaller 128-probe budget anchors: 4–8 on OpenAI, 64 on Vogue, 100 on arXiv (FULL later selects 200). These are workload associations, not causally isolated or matched-frontier parameters.

The interface, memory and compilation differences in section 3.3 remain relevant to small margins. Section 7.8 examines their empirical sensitivity.

7.2.5 Small batches change the competitive region

T5 compares scheduled JHQ, CAGRA fp32/int8 and IVF-RaBitQ at batches 8/32 on the four E1 datasets. Five synchronous host-to-host rounds use 1,000/992 queries, respectively. Its 1,808 points include JHQ's one-block control; IVF-PQ and brute force are outside this campaign. Figure 7.3 and table 7.3 compare the resulting regions.

At recall 0.97 and batch 8, CAGRA fp32 leads on all four datasets under both estimators: interpolated JHQ ratios are 0.641–0.893. At batch 32, these ratios are 0.622–1.008, with OpenAI-3072 close under interpolation but favourable to JHQ under the measured threshold.

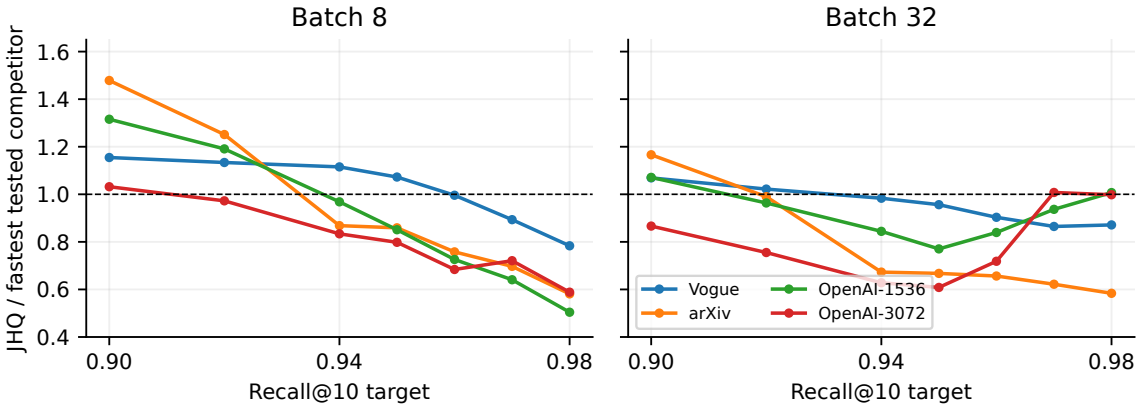


Figure 7.3. T5 JHQ / fastest-competitor QPS under interpolation with qualifying measured endpoints. Above one favours JHQ. Strict measured thresholds are in table 7.3.

Table 7.3. T5 QPS ratios: interpolation with qualifying endpoints / strict measured threshold. The fastest competitor is selected independently for each estimator; higher-recall qualifying endpoints remain eligible without extrapolation.

Dataset	b	$R = 0.90$	$R = 0.95$	$R = 0.97$
Vogue	8	1.155 / 1.091	1.073 / 0.980	0.893 / 0.854
Vogue	32	1.069 / 0.997	0.956 / 0.895	0.865 / 0.839
arXiv	8	1.479 / 1.454	0.859 / 0.851	0.697 / 0.698
arXiv	32	1.166 / 1.148	0.667 / 0.661	0.622 / 0.582
OpenAI-1536	8	1.316 / 1.131	0.851 / 0.825	0.641 / 0.620
OpenAI-1536	32	1.071 / 0.915	0.770 / 0.752	0.937 / 0.914
OpenAI-3072	8	1.032 / 0.953	0.798 / 0.798	0.721 / 0.702
OpenAI-3072	32	0.866 / 0.734	0.608 / 0.771	1.008 / 1.351

At recall 0.95, Vogue's batch-8 lead likewise appears only under interpolation. Grid resolution therefore matters to borderline rankings.

Lower-recall comparisons must retain competitors whose fastest frontier point already exceeds the target. At recall 0.90, batch-8/32 ratios are 1.032–1.479/0.866–1.166; strict thresholds give three/one JHQ wins. Omitting qualifying endpoints would overstate these low-recall leads. Scheduling gains 1.7–3.3× over one block at recall 0.95, without guaranteeing a cross-system win.

Null-call medians are 0.012–0.072 ms and remain included. CAGRA uses Python, JHQ/RaBitQ C++; warm-ups differ (section A.5).

7.3 Cartesian LUT, Packed Loads and Lane Refinement

The RQ2 ablations measure both kernel and application effects.

7.3.1 Exact table factorization and its throughput consequence

The LUT ablation tests the cost model in section 5.2 and FP32 ranking effects.

Figure 7.4 compares Cartesian table grouping on six datasets, probe counts 8/32/128/512, fixed $\alpha = 100$, and $g_{\text{LUT}} \in \{1, 2, 4, 8\}$. Five rounds use batch 1,000 (Stella 999). Two groups lead in 22/24 cells. Exceptions are near ties: arXiv/128 favours one group by 0.23%; OpenAI-3072/8 favours four by 0.84%. Paired two-versus-one wins are 23/24; all dataset geometric means exceed one (1.021–1.318). Shared-index cells are not independent trials.

The two-group/one-group ratio ranges from 0.998 to 1.914. Four and eight groups each store 16 entries but require more lookups and additions. These arms show that smaller tables need not improve throughput; occupancy and instruction costs were not isolated. The maximum



Figure 7.4. Cartesian table grouping at fixed $\alpha = 100$ (v71). Group counts describe g_{LUT}, not parallel scan groups. Curves are medians of per-dataset QPS ratios to the one-group arm; shading shows their range.

Table 7.4. Refinement at 128 probes. Ratios are paired end-to-end gains over L0, except the last column, which is the profiled refinement-kernel gain. The kernel and application gains must not be interchanged.

Dataset	α	$L1/L0$ QPS	$L16/L0$ QPS	Kernel gain
arXiv	100	1.368	1.789	5.97
BGE	32	1.026	1.062	1.97
OpenAI-1536	8	1.055	1.145	3.34
OpenAI-3072	4	1.063	1.171	8.81
Stella	16	1.030	1.053	2.66
Vogue	64	1.161	1.240	3.72

reported recall spread within any four-arm cell is 10^{-4}. At 128 probes and $\alpha = 100$, T4 compares identical candidates on Vogue, arXiv and OpenAI-3072. Full versus $16 + 16$ tables differ in 22.6–26.1% of candidate-score bit patterns, with maximum relative error 4.53×10^{-7}. Across all forms, 20.5–29.4% differ, with maximum relative error 4.64×10^{-7}. Vogue has 2–4/256 final-ID differences, while arXiv/OpenAI have none. Unchanged primary survivor sets can still produce different final IDs because the rounded primary score enters refinement (section A.5).

7.3.2 Packed access and cooperative residual scoring

Packing wins 1.012–1.658× across 48 matched pairs (six datasets, four probe counts, $g_{LUT} \in \{1, 2\}$), with recall differences at most 10^{-4}. Both layouts are subspace-major. This tests packing/load/unpack together; one aggregate rate per arm gives less timing evidence than v71's five rounds.

Figure 7.5 and table 7.4 then separate wider loads from lane cooperation in residual refinement. L0 is the original scalar path; L1 changes the load path without distributing a candidate across multiple lanes; L16 assigns 16 lanes to a candidate. The same process rotates arm order across five timing rounds, with three search calls per arm per round. For the twelve ordinary configurations, L16 improves end-to-end throughput by 1.015–1.789×. An additional arXiv configuration at $\alpha = 200$, explicitly a high-budget control, reaches 2.383×; it is not the gain range of the ordinary selected budgets.

The load-only control is bit-identical to L0 in the tested cases, yet already improves arXiv throughput by 1.368×. Lane cooperation adds a larger reduction in refinement time. On arXiv at $\alpha = 100$, the profiled refinement kernel falls from 4,277.5 to 716.8 μs, a 5.97× gain, while

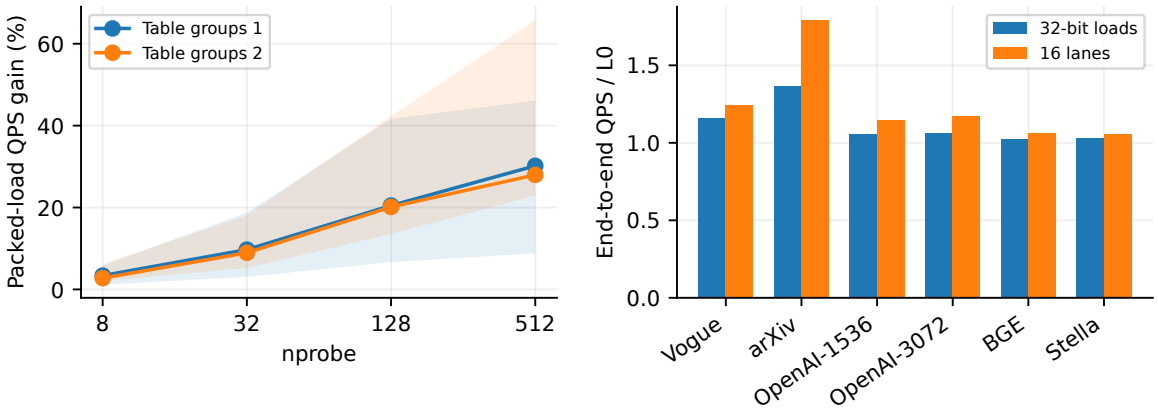


Figure 7.5. Packing and refinement controls. Left: median packed-load QPS gain across six datasets, with shaded minimum–maximum ranges, for each table grouping and probe count. Right: measured load-only and 16-lane end-to-end gains at 128 probes and the ordinary dataset-specific budgets. These panels come from different experiment families.

total throughput improves 1.789×. OpenAI-3072's refinement kernel improves 8.81×, but total throughput improves 1.171×. The remaining routing, screening, selection and transfer costs limit the application gain.

Using refinement's baseline wall-time fraction f and kernel speedup s, Amdahl's estimate is $1/(1-f+f/s)$. For arXiv/OpenAI-3072, $f = 0.529/0.168$ predicts 1.787/1.175×, versus 1.789/1.171× measured. Separate profiling makes these explanatory estimates.

Lane reductions change FP32 association: reported relative distance error reaches about 1.6×10^{-6}, with occasional ranking changes. Sanitizer checks cover small recorded subsets; scoring is not generally bit-identical.

7.4 Global Top-c, Shared Bound and Probe Correctness/Ablation

The representation checks in section 4.6 concern the stored index; this section evaluates query selection and its cost.

7.4.1 Strict CPU reference for computed-score selection

T4 CPU-sorts GPU-produced primary scores under $(D_{\mathrm{P}}, \text{position})$. Its exporter follows scan layout/accumulation and matches returned valid scores bitwise. Source comparison confirms that the diagnostic version changes only export, relative to the table-sweep implementation. The 270 configurations cover Vogue/arXiv/OpenAI-3072, 1/8/128 probes, $\alpha \in \{4, 100, 200\}$, $g_{\mathrm{scan}} \in \{0, 1, 2, 4, 8\}$ and both bound settings: 69,120 query instances (table 7.5).

Table 7.5. T4 CPU-reference checks. Set and order equality use the complete primary key. Valid count checks $\min(c, n_c)$ survivors; final equality sorts retained computed refined scores by $(D_{\mathrm{H}}, \mathrm{ID})$, including their FP32 bits. Counts are query instances, with queries reused across configurations.

Path	Queries	Set / order exact	Tie-only	Valid count	Final exact
$g_{\mathrm{scan}} \geq 1$	55,296	55,296 / 55,296	0	55,296	55,296
$g_{\mathrm{scan}} = 0$	13,824	13,768 / 8,758	56	13,824	13,824

All 55,296 grouped instances match complete-key order. All 13,824 one-block instances satisfy distance order, with 56 cutoff-tie set differences. Underfull valid counts pass in 19,120 cases; all 69,120 final selections match retained-score references. These test lemma 5.1

Table 7.6. v69/v65 end-to-end throughput ratios for the exact computed-score probe change. Each workload–batch comparison has ten timing rounds per binary, collected in alternating runs. Values above one favour v69.

Dataset	Batch 8	32	256	1,000
arXiv	1.074	1.063	1.035	1.034
Vogue	1.109	1.103	1.052	1.041
OpenAI-3072	1.167	1.117	1.055	1.050

and propositions 5.1 and 5.2; separate scan/bound and sanitizer checks have their own coverage (section A.5).

7.4.2 Shared bounds preserve the selection target

Proposition 5.2 establishes the selection guarantee. The paired experiment measures the work saved by sharing that bound.

All 48 paired bound comparisons agree on primary/final output; throughput spans 0.994–1.562×. At arXiv $\alpha = 200$, batch 1,000 and eight groups, sharing cuts merge 2,629→471 μs and scan 7,968→5,513 μs. Other cells include slight regressions.

The pilot's first group holds 46–81% of winners. Equal quotas need fallback; unequal quotas leave a dominant group. Shared bounds avoid that fixed-quota choice.

7.4.3 Exact probe selection and the limits of pipeline identity

The radix probe selector keeps prior score arithmetic, including OpenAI TF32 (section 7.2.3). Its 1,792 synthetic cases pass. On all 1,000 arXiv/Vogue/OpenAI-3072 queries, probes and primary positions/distances match the previous selector.

Table 7.6 shows end-to-end improvements of 1.034–1.167×. At batch 8, the profiled probe stage falls from 56.5 to 22.6 μs on arXiv, 52.7 to 14.7 μs on Vogue, and 77.8 to 16.7 μs on OpenAI-3072. The unchanged stages limit full-batch gains. These within-family ratios have different baselines from the preceding ablations.

On Vogue, the probe/primary dump is byte-identical across the two binaries for all 1,000 queries in the one-group, batch-8 test, yet final recall is 0.9642 versus 0.9644. The same baseline binary also changes from 0.9642 in the identity run to 0.9644 in each of its two timing runs. The variation therefore also occurs without the probe change and lies after primary selection in the dump-controlled case. Its root cause remains unresolved. Scheduled and one-block outputs are bit-identical within each checked process; dump coverage is in appendix A.

Finally, the optional v70 FP16 experiment is a separate approximation. Its probe sets match the v69 reference on 936/1,000 arXiv queries, 979/1,000 Vogue queries and 998/1,000 OpenAI-3072 queries, with very high mean set overlap. High overlap and small recall changes do not make the primitive exact. Those results are excluded from the main E1 frontier and from the exact-probe claim.

7.5 Held-out Budget Quality, Throughput and Payback

RQ3 compares the policy of chapter 6 with fixed $\alpha = 100$ and the stronger universal control $\alpha = 64$.

7.5.1 Unlabelled selection and a labelled diagnostic oracle

The six datasets are tested at 128 and 512 probes with the fixed, sampled, FULL, enumeration and oracle controls in section 6.4. The disjoint pools and 64 draws per sample size follow section 3.2.

All rule variants use equation (6.4) with $k = 10$ and $\epsilon = 1$. FULL uses all calibration queries; the diagnostic oracle uses held-out labels (table 6.1).

Table 7.7. Full-pool calibration ($S = 500$). Loss is relative to the best measured held-out recall; rates are kQPS from the selected fixed arm. "Cal." is one full-pool calibration call per cell, not a repeated stream timing. Held-out counts are 500, or 499 for Stella.

Dataset	Probe	α_{full}	Loss	Q_{full}	vs 100	vs oracle	Cal. ms
Vogue	128	100	0.0000	202.9	1.000	0.908	10.1
arXiv	128	200	0.0000	151.5	0.728	1.000	10.0
OpenAI-1536	128	8	0.0000	267.7	1.599	1.000	9.5
OpenAI-3072	128	4	0.0000	145.0	1.691	1.000	17.1
BGE	128	64	0.0006	124.9	1.075	0.932	17.1
Stella	128	16	0.0006	110.4	1.153	1.000	20.6
Vogue	512	100	0.0002	100.1	1.000	0.969	20.9
arXiv	512	200	0.0000	69.0	0.831	1.000	25.1
OpenAI-1536	512	8	0.0000	114.9	1.326	1.000	21.8
OpenAI-3072	512	4	0.0002	52.1	1.219	0.977	40.5
BGE	512	64	0.0006	50.8	1.033	1.000	42.3
Stella	512	16	0.0006	36.0	1.041	1.000	60.3

Table 7.8. Held-out quality at $S = 128$. Rule α is the median across 64 draws; oracle α uses held-out labels. $\overline{R}_{\text{rule}}$ is reconstructed mean recall. "Losses" counts draws with recall loss $> 10^{-3}$ relative to the best tested held-out recall; L_{95} is the logged lower-order 95th-percentile loss estimate. R_{100} is fixed-budget recall.

Dataset	Probe	α: rule/oracle	R_{100}	$\overline{R}_{\text{rule}}$	Losses	L_{95}
Vogue	128	64 / 64	0.9680	0.9677	5/64	0.0022
arXiv	128	100 / 200	0.9748	0.9751	40/64	0.0024
OpenAI-1536	128	8 / 8	0.9280	0.9279	4/64	0.0000
OpenAI-3072	128	4 / 4	0.9252	0.9252	0/64	0.0000
BGE	128	32 / 32	0.9198	0.9190	1/64	0.0008
Stella	128	16 / 16	0.9892	0.9884	6/64	0.0030
Vogue	512	64 / 64	0.9916	0.9915	2/64	0.0002
arXiv	512	200 / 200	0.9878	0.9885	32/64	0.0016
OpenAI-1536	512	8 / 8	0.9740	0.9738	4/64	0.0000
OpenAI-3072	512	4 / 8	0.9704	0.9702	0/64	0.0002
BGE	512	64 / 64	0.9670	0.9661	24/64	0.0014
Stella	512	16 / 16	0.9954	0.9945	7/64	0.0030

7.5.2 Full-pool calibration as the quality-first working point

Full-pool calibration uses all 500 calibration queries once per cell. Its selected arms meet the held-out 10^{-3} loss target in 12/12 cells; the largest observed loss is 0.0006. Their rates are 0.908–1.000 of the GT oracle's, and calibration costs 9.5–60.3 ms. The geometric-mean rate ratio to fixed 100 is 1.111×, or 1.193× over the ten cells where fixed 100 also meets the target. Table 7.7 shows the complete comparison. Against fixed 64, full-pool selection is faster in six cells, equal in two and slower in four: its main advantage is the observed coverage of all twelve quality targets.

These cells favour FULL for quality coverage and $S = 128$ for lower calibration cost, within the statistical scope of section 6.2.

7.5.3 Lower-cost sampling and its quality risk

Table 7.8 reports the $S = 128$ result. The median selected budget matches the oracle budget in ten of twelve cells; it differs on arXiv at 128 probes and OpenAI-3072 at 512. The selected budgets span 4–200, indicating substantial workload dependence. Median budget and mean recall need not describe one arm. The latter is reconstructed from printed best recall minus mean loss and is approximate at four decimals.

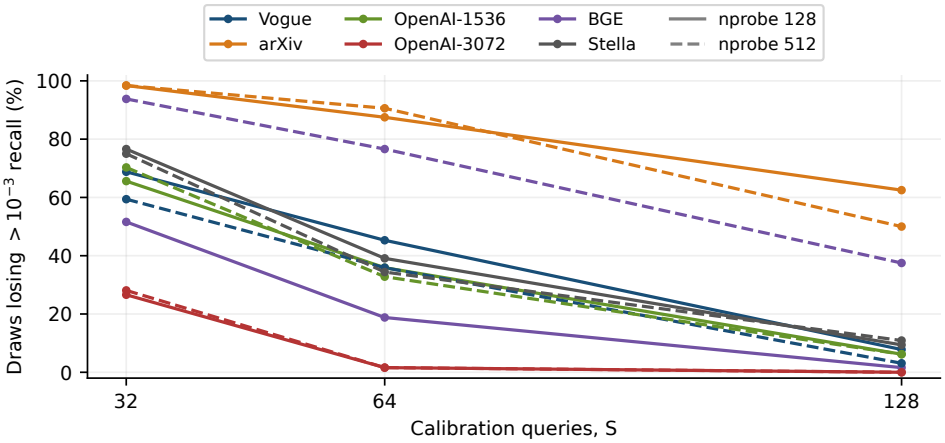


Figure 7.6. Held-out risk across sample sizes. The vertical axis is the percentage of calibration draws losing more than 10^{-3} held-out recall relative to the best tested budget. Increasing S also tightens the allowed slot-disagreement fraction because equation (6.4) keeps threshold one.

Table 7.9. Calibration economics, $S = 128$. Rates are kQPS and rule rate is mean selected-arm QPS. Ratios compare with fixed 100 and fixed 64. The "Batches" column gives payback relative to fixed 100 using 500 queries, or 499 for Stella; "Never" denotes no throughput payback. The quality column separately marks whether fixed 100 meets the held-out loss target.

Dataset	Probe	Q_{100}	$\overline{Q}_{\text{rule}}$	vs 100	vs 64	Cal. ms	Batches	Fixed 100 quality
Vogue	128	202.9	224.4	1.106	1.004	4.0	17.0	Pass
arXiv	128	207.9	188.8	0.908	0.821	4.2	Never	Fail
OpenAI-1536	128	167.5	268.5	1.603	1.367	3.9	3.5	Pass
OpenAI-3072	128	85.7	145.0	1.691	1.230	5.7	2.4	Pass
BGE	128	116.2	131.2	1.130	1.051	6.5	13.2	Pass
Stella	128	95.7	110.3	1.153	1.087	7.5	10.8	Pass
Vogue	512	100.1	102.9	1.028	0.996	7.5	55.4	Pass
arXiv	512	83.1	76.2	0.917	0.870	9.7	Never	Fail
OpenAI-1536	512	86.6	115.0	1.327	1.268	8.8	6.2	Pass
OpenAI-3072	512	42.8	52.1	1.219	1.052	13.1	6.2	Pass
BGE	512	49.2	51.1	1.039	1.006	15.8	41.2	Pass
Stella	512	34.6	36.0	1.041	1.028	21.3	37.1	Pass

ArXiv has 40/64 and 32/64 loss crossings at 128/512 probes, despite mean loss 0.0009. Fixed 100 loses 0.0012/0.0016: better than default need not meet the target. BGE/512 has 24 crossings and $L_{95} = 0.0014$; Stella reaches 0.0030; OpenAI-3072 has none (table 7.8). Four losses can coexist with reported $L_{95} = 0$ (section A.3).

7.5.4 Estimated throughput and calibration economics

Using the mean selected-arm QPS estimate specified in section A.3, $S = 128$ improves over fixed $\alpha = 100$ in ten cells by 1.028–1.691×. On arXiv the ratios are 0.908 and 0.917: mean recall improves, but throughput falls, giving no throughput payback.

Against fixed 64, sampled-arm QPS wins 9/12 cells: geometric mean 1.054×, range 0.821–1.367× (OpenAI 1.052–1.367×). Both fixed controls fail arXiv quality, so these are not uniformly quality-matched gains.

Calibration takes 3.9–21.3 ms. Under section 6.5's reuse assumptions, equation (6.15) and table 7.9 give maximum positive paybacks of 0.54/2.72 s versus fixed 100/64; FULL gives 1.52/2.25 s. ArXiv controls fail quality. Direct reuse below checks total cost and recall together.

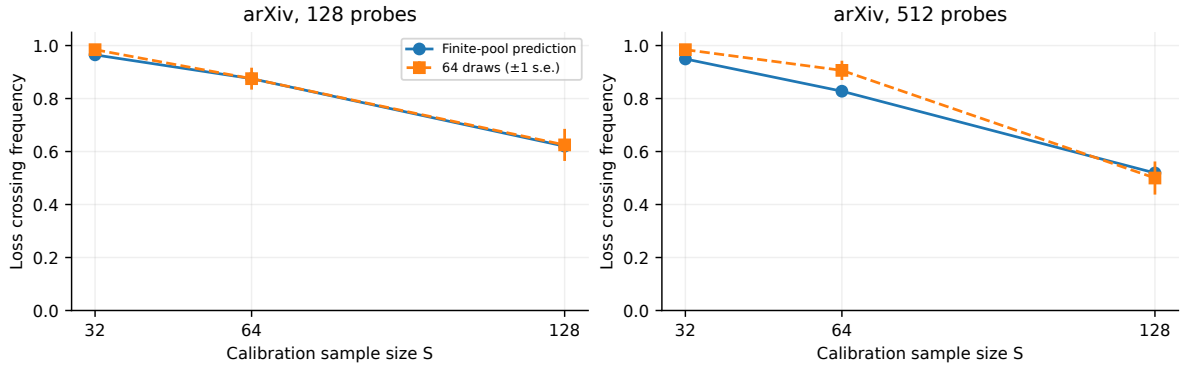


Figure 7.7. Finite-pool prediction and observed arXiv loss crossings. Bars show one estimated standard error over 64 independently drawn subsets conditional on the fixed pool, not uncertainty over future workloads.

7.5.5 Complete decision traces and finite-pool risk

All 2,304 sampled and twelve FULL curves are monotone and match enumeration. Recorded subset sums reproduce 18,432 grid counts (equation (6.9)). This checks those aggregates, not arbitrary subset/position invariance or the distance-only cases' general nesting (section 6.3).

Maximum sampled loss reaches 0.0050 (arXiv/512) and 0.0046 (BGE/512) at $S = 128$, and 0.0294 at $S = 32$. FULL stays within 0.0006.

ArXiv/128 predicted/observed crossing rates are 0.965/0.984, 0.875/0.875 and 0.620/0.625 for $S = 32, 64, 128$ (figure 7.7). Across 36 comparisons, the largest gap is BGE/512's 0.144 (0.519/0.375). Prediction assumes stable counts and uses held-out fixed-arm quality to map choices to GT loss; calibration remains label-free.

Ten alternating repeats support a low-budget timing plateau, without implying generally monotone runtime.

7.5.6 Direct calibration-plus-reuse measurement

Four cells run fixed 64/100, sampled and FULL policies over five rotated rounds of 500 repeated held-out batches. Timing includes gathering, calibration, budget installation and first-batch graph setup; build, warm-up and scan calibration are excluded (table 7.10).

Table 7.10. Direct total milliseconds for 500 reuse batches, averaged over five rounds, including budget calibration for sampled and FULL policies. Each batch has 500 queries. Quality is discussed with the timing results.

Dataset/probes	Fixed 100	Fixed 64	Rule $S = 128$	FULL
arXiv/128	1224	1102	1291	1680
OpenAI-1536/128	1491	1264	852	860
OpenAI-3072/128	2757	2160	1769	1783
BGE/512	5185	5018	4921	5062

OpenAI-3072/128 always selects 4: sampling takes 1,769 versus fixed-100's 2,757 ms, both at recall 0.9252. FULL selects the same budget and takes 1,783 ms including its larger calibration cost. OpenAI-1536 FULL gives 860 versus 1,491 ms at recall 0.9280; sampling gives 852 ms at 0.9276.

Direct cumulative savings begin at 4/6 batches (OpenAI-1536) and 4/9 (OpenAI-3072) versus fixed 100/64 (figure 7.8); fractional fits are separate estimates. BGE sampling gives 4,921 versus 5,185 ms but recall 0.9658 versus 0.9670; FULL gives 5,062 ms and 0.9664. ArXiv FULL selects slower 200 for quality; fixed controls fail the target.

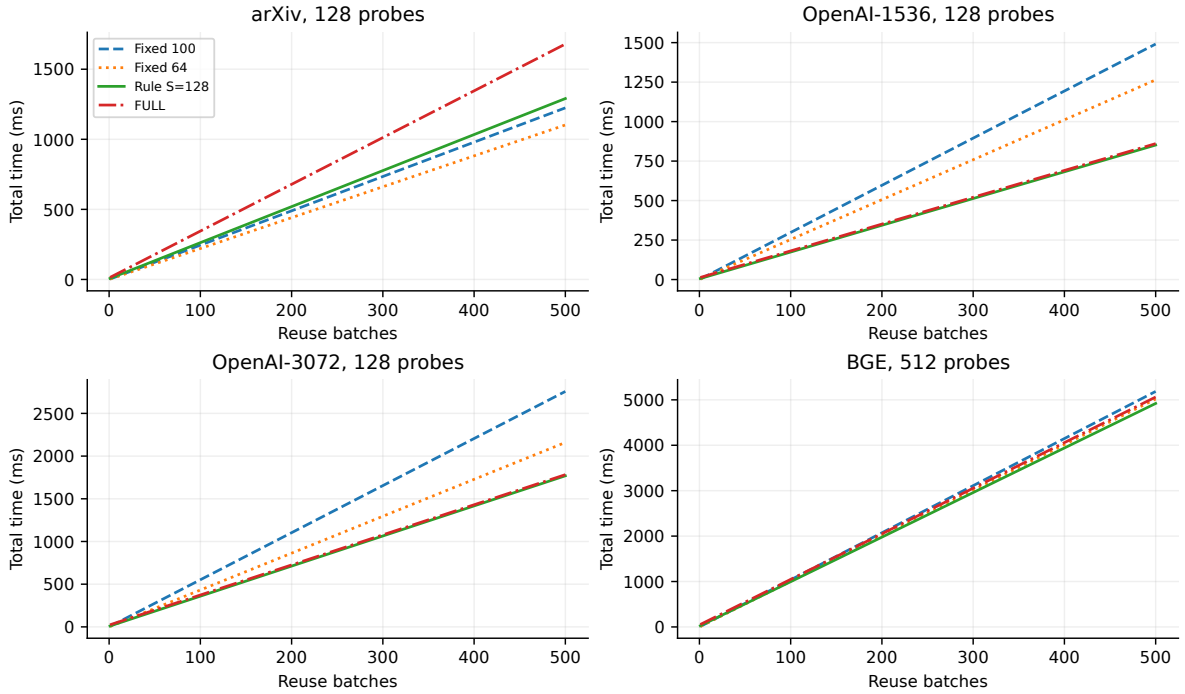


Figure 7.8. Direct mean cumulative time, including budget calibration and first-batch setup. Each curve replays one fixed held-out query pool. Policies can have different recall; timings alone are not a quality-matched ranking. Dataset/probe settings match table 7.10.

Table 7.11. Integrated execution gains over v61 L16. Entries are geometric means of per-cell median paired speedups from five rounds. "Held out" excludes the four configurations used to set the rule; it means held-out configurations, not the budget experiment's held-out query split.

Batch	Cells	$g_s = 2$	$g_s = 4$	$g_s = 8$	Scheduled	Held out
8	13	1.318	2.014	2.728	2.727	2.930
32	13	1.323	1.970	2.337	2.356	2.560
256	13	0.997	1.090	1.099	1.057	1.041
1000	11	0.959	0.947	0.886	1.015	1.012

7.6 Batch Regimes, G Scheduling and Workload Skew

This section answers the performance-regime part of RQ2: when additional scan groups supply useful parallelism, and when their extra work dominates.

7.6.1 Parallelism versus repeated selection work

At batch 8, eight groups raise throughput 2.06–2.76×: arXiv scan drops 849→216 μs, with 16.7 μs merging. At batch 1,000 and $\alpha = 200$, independent grouping instead raises scan 3,530→7,652 μs plus 2,624 μs merging, reaching only 0.46× one-block QPS. Shared bounds reduce work, without removing this regime change.

7.6.2 A fixed rule evaluated on further configurations

The scheduler follows the fill/skew rule and pseudoquery calibration in section 5.5. Four configurations set the constants before the additional configurations were measured.

The means use thirteen small-batch configurations and eleven full-batch ones (Stella has 999 queries); held-out subsets use nine/seven (table 7.11). Fixed eight groups fall to 0.886× at batch 1,000. Worst scheduling regret is 1.217 on BGE/batch 256.

Vogue's CV 0.338–0.622 forces four groups at batch 1,000. Scheduled/one-block QPS spans 0.895–1.175; $(n_{\text{probe}}, \alpha) = (16, 16), (16, 64), (32, 64)$ lose at 0.955, 0.895, 0.955. CV captures

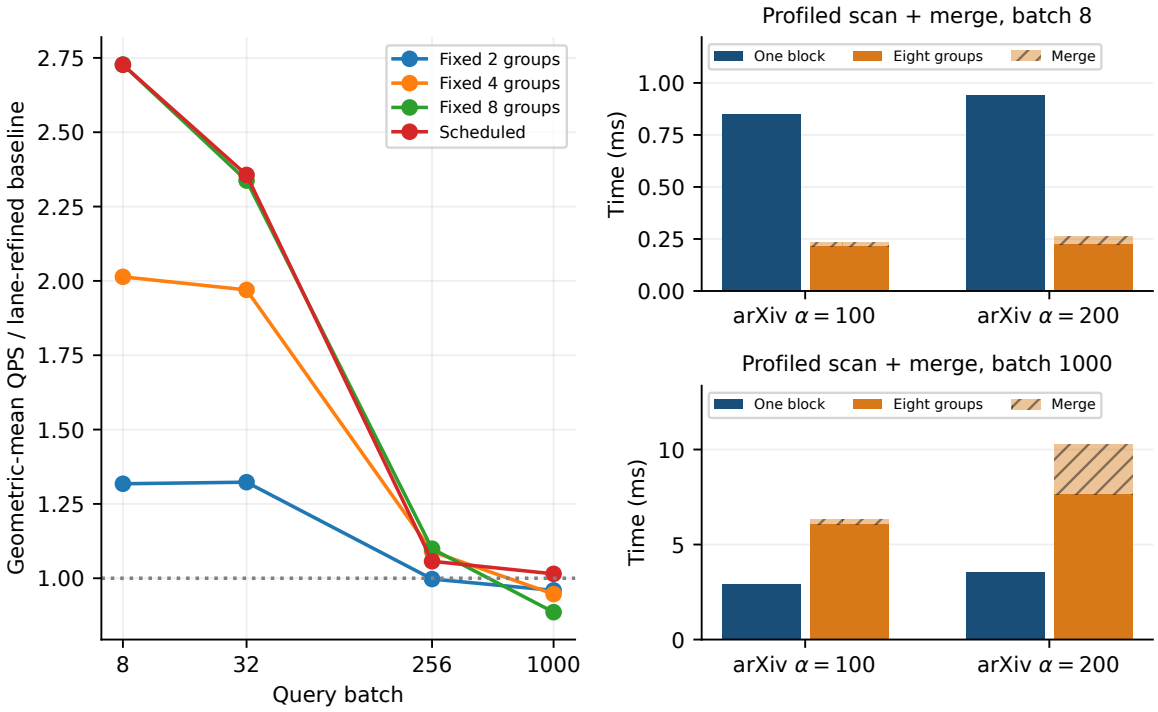


Figure 7.9. Execution regimes. Left: fixed shared-bound group counts and scheduled execution relative to the lane-refined one-block baseline (13 cells at small batches, 11 at batch 1,000). Right: paired scan and merge profiles at batches 8 and 1,000 for arXiv budgets 100 and 200. Hatching denotes merge time. The right panels use the independent-bound scan experiment; they explain a cost, not the scheduler's entire speedup.

dispersion, not absolute work: the same rule enables the 0.97 lead but wastes work at low probes (section 7.2).

Across 250 variant/configuration pairs and 234,320 query comparisons, there are no non-tie primary differences or final-ID-set differences. Tied output order can differ. These are sequential synchronous throughput tests, not open-loop tail-latency tests.

7.7 Cold Construction, Resident Memory and Tested Scale

RQ1 is assessed through cold construction, resident memory and completed large-dataset searches; the E1 memory comparison also informs RQ4.

7.7.1 Cold builds and the lifecycle boundary

Three empty-cache repeats give Vogue training 298–303 ms and total about 0.56 s; its archived 1.33 s run was slow. Stella takes 10.54–10.65 s, near the archived 10.40 s (table 7.12).

Construction and startup boundaries follow sections A.2 and 3.3; figure 7.10 compares these repetitions with baseline configurations.

Stella builds take 74.55/82.45 s for isolated CAGRA int8 (including quantisation) and a three-configuration median 86.59 s for IVF-PQ, versus JHQ's 10.54–10.65 s. Quality and resident memory remain separate comparisons.

Five builds show up to 0.0012 recall range on arXiv. Stella startup is 18.2–18.6 s with warm pages, 129.9 s on first read. At 20 ms resolution, peaks are BGE 26.08 GiB and Stella 22.58 GiB. BGE's 12 GiB assignment buffers precede its 23.05 GiB gather payload; summing them would overstate simultaneous memory (equation (4.13)).

Table 7.12. Cold JHQ construction and separately measured resident memory. Train/add are component medians of three supplementary normal builds; Total is the median paired sum. Memory comes from v71's two-group, 128-probe, $\alpha = 100$ runs and includes context and workspace. "Residual" is the modelled residual-code share of the measured used total, not the fraction read per query.

Dataset	Train s	Add s	Total s	Used GiB	Raw GiB	Residual
Vogue	0.30	0.26	0.56	1.47	2.67	45.4%
arXiv	0.48	0.51	0.99	2.58	6.45	62.5%
OpenAI-1536	0.61	0.38	1.00	2.39	5.72	59.7%
OpenAI-3072	1.16	0.67	1.82	4.14	11.43	69.0%
BGE	3.56	3.65	7.20	11.93	38.50	80.7%
Stella	3.55	6.99	10.55	20.44	67.81	82.9%

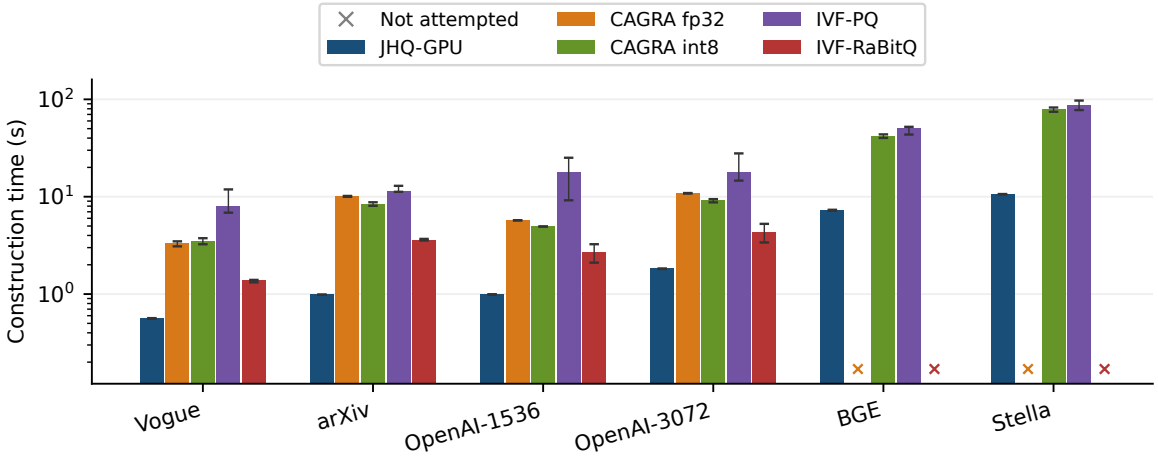


Figure 7.10. Recorded construction costs, with CAGRA int8 included. JHQ is the median and range of three supplementary normal cold builds with warm input pages; other bars are medians across successful measured index configurations, with minimum–maximum whiskers. Int8 includes dataset quantization; RaBitQ excludes the required serialization round trip. Missing large-dataset bars denote unattempted configurations, not measured build failures. The vertical axis is logarithmic.

7.7.2 Resident allocation and where compression helps

Figure 7.11 combines the component model in section 4.4 with measured resident totals. Its unattributed remainder is retained, and the preceding construction measurements give the sampled peaks.

Residuals occupy 80.7% of BGE's 11.93 GiB and 82.9% of Stella's 20.44 GiB totals, including context/workspace. Raw vectors need 38.50/67.81 GiB. Smaller α saves reads, not stored residual bytes (section 4.4).

The E1 measurements provide a separate cross-system view in table 7.13. Subtracting the baseline-measured 507.2 MiB context from JHQ's process total gives approximately 0.29–0.39 of CAGRA fp32's index allocation. JHQ's figure includes workspace, whereas CAGRA's is the build-time index delta. These are parameter-sweep ranges, not memory-matched recall points or a comparison of equal peak budgets. The CAGRA int8 allocations in the same table are of a similar order to JHQ's: compression relative to fp32 is shared by competing designs. The component model in table 4.1 explains this comparison.

7.7.3 Large-dataset feasibility and competitive regions

BGE contains 10,091,524 vectors and Stella 17,776,615, both with 1,024 dimensions and 32,768 IVF lists. Stella's queries are sampled from the corpus; its result therefore characterizes

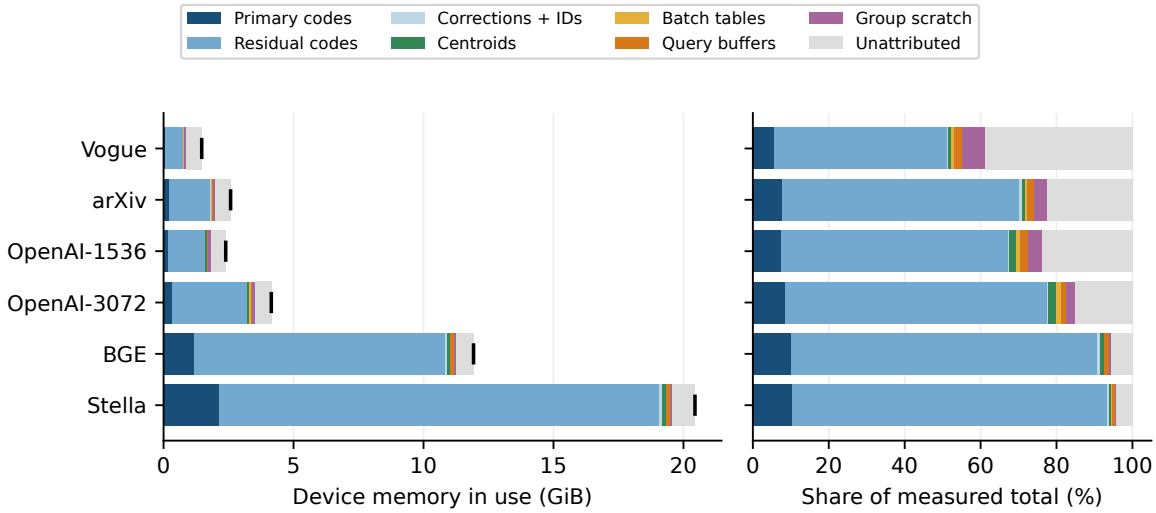


Figure 7.11. Resident-memory decomposition at two-group tables, 128 probes and $\alpha = 100$. Component sizes are modelled, markers are measured used totals, and hatching is the unattributed remainder. The totals include context and workspace. The preceding construction measurements give sampled peaks; these resident markers are not peaks.

Table 7.13. E1 allocation ranges. JHQ is process total minus a 507.2 MiB context estimate, including query workspace; both CAGRA columns are index allocations measured during build. Ranges cover the parameter sweep.

Dataset	JHQ net process GiB	CAGRA fp32 index GiB	CAGRA int8 index GiB
Vogue	0.90–1.01	2.89–2.91	0.89–0.91
arXiv	2.01–2.19	6.84–6.98	2.01–2.15
OpenAI-1536	1.79–2.33	5.95–5.97	1.67–1.68
OpenAI-3072	3.54–3.82	11.67–11.68	3.10–3.11

that benchmark rather than an independently collected external-query distribution. Figure 7.12 includes successful CAGRA int8 and IVF-PQ measurements. Full-precision CAGRA, RaBitQ configurations and FP32 brute force were not attempted in the large-scale comparison when their computed storage requirements exceeded the available device capacity. Those omissions are planned feasibility exclusions, not observed out-of-memory errors or measured recall failures.

CAGRA int8 fits Stella. On BGE it is 1.35–2.96× faster at recall 0.88–0.95; JHQ is 1.086× faster at int8's maximum 0.9579. Maximum BGE/Stella recalls are int8 0.9579/0.9783, JHQ 0.9663/0.9952 and PQ 0.9525/0.9896.

On Stella, JHQ gives 185.4 kQPS at recall 0.978679; int8 gives 66.3 kQPS at 0.978278. That 2.80× measured-point ratio uses different recalls. Int8 has no coverage for much of JHQ's upper region (appendix A).

7.8 Numerical Robustness, Anomalies and Recall Attribution

The robustness checks qualify the RQ2 exactness claims, the RQ3 risk assessment and the small cross-system margins in RQ4.

7.8.1 Quality resolution and repeated execution

One GT hit changes Recall@10 by 10^{-4} over 1,000 queries or 2×10^{-4} over 500 (slightly more over Stella's 499). This resolution is not an uncertainty interval.

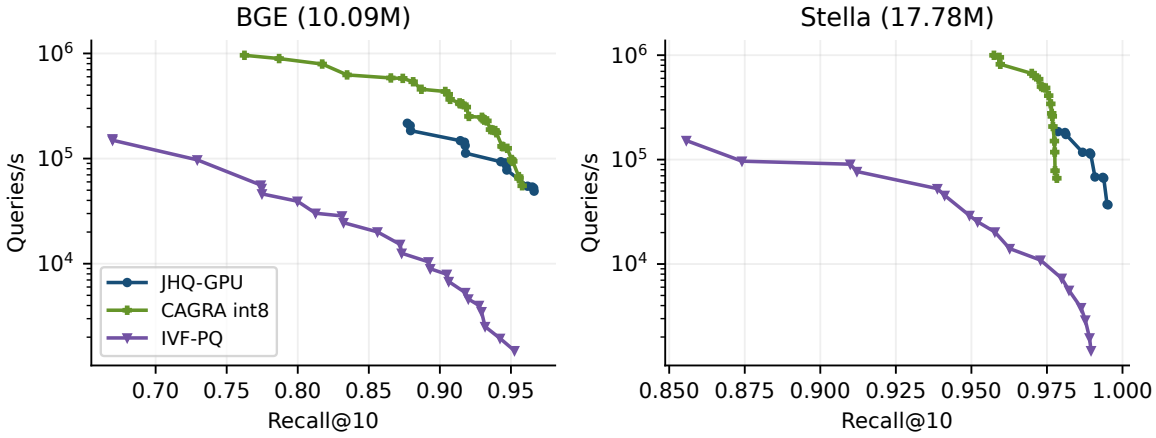


Figure 7.12. Large-dataset host-to-host frontiers, including the successful isolated CAGRA int8 reruns. BGE uses 1,000 queries and Stella 999. An endpoint is the end of the tested grid, not an intrinsic recall limit.

Repeated calls and repeated index builds answer different questions. E1 retains 369 baseline repeat-call identifier flags, including tie-order changes; these points remain in the frontiers. The 401-count convention and cross-protocol checks are separated in section A.2.

Near 0.97, local frontier slopes imply 1.01–3.17% QPS change per 10^{-3} recall before snapping. Five-build recall ranges are Vogue 0.0008, arXiv 0.0012, OpenAI-1536 0.0004 and OpenAI-3072 0.0002. They suggest few-percent local sensitivity, not a bound from retraining whole frontiers/baselines. FP32 sensitivity retains the OpenAI advantage within its cached-model scope (section 7.2.3); arXiv and LUT near ties remain inconclusive.

RaBitQ's median five-round ranges are 3.79%, 3.34%, 2.87% and 5.25% of median QPS on Vogue, arXiv and the two OpenAI datasets. These are descriptive spreads, not ratio confidence intervals. Traces include occasional 30–60 ms calls. As a sensitivity check, recomputing QPS as $10^6/\,\text{median(call latency in ms)}$ for a 1,000-query call reduces the influence of long calls. At recall 0.97, the resulting JHQ/RaBitQ ratios are 1.063, 0.995, 1.118 and 1.100. The three favourable directions and arXiv near parity remain, so the result is not solely a consequence of long calls.

7.8.2 Exactness is attached to a particular stage

The stage contracts in table 5.1 distinguish computed-score selection from raw-space kNN; lane rounding, cutoff ties and process variation qualify different stages.

The archived reconstruction pilot measures the remaining gap to full-probe reconstructed scoring. Across seven workload–budget arms of 128 queries each, scoring all probed candidates in FP64 improves recall over GPU outputs at selected budgets by 0–0.002344. For arXiv at 128 probes, $\alpha = 200$ matches the full-IVF reconstructed answer set on all 128 queries. At 100, the recall is 0.9734375 versus 0.9750 for full-IVF scoring: an additional 0.0015625 budget deficit beside the joint routing/reconstruction deficit of 0.0250 (section 6.2). These are pilot-specific aggregate differences; the seven arms include two budgets on one dataset–probe workload. FP64 reconstructed scoring also differs from the GPU arithmetic, so the pilot does not isolate a pure budget effect in every arm. Separating routing from representation requires additional stage-wise GT coverage.

8 Discussion and Future Work

8.1 Three Cross-Layer Insights

8.1.1 A useful representation exposes a useful unit of reuse

Mailman loses despite more reuse; two-group LUTs usually win (figure 7.4). Reuse must fit table placement and access (sections 5.2 and 5.3), so codebook quality and attainable scorer cost need joint evaluation.

8.1.2 Useful parallelism is bounded by duplicated state and selection

Once query parallelism is sufficient, staging/selection/merge overhead can dominate (figure 7.9). Vogue and BGE expose missing absolute-work and decomposition features (sections 7.2 and 7.6). A richer scheduler must also repay calibration; integrated results do not isolate shared bounds.

8.1.3 Execution speed changes the value of avoiding refinement

Consider a latency model with fixed non-refinement cost $T_0 > 0$, high-/low-budget refinement times $F_h \geq F_\ell > 0$, and a common refinement-time multiplier $0 < \eta \leq 1$. The low-budget speedup and its derivatives are

$$S(\eta, T_0) = \frac{T_0 + \eta F_h}{T_0 + \eta F_\ell}, \tag{8.1}$$
$$\frac{\partial S}{\partial \eta} = \frac{T_0(F_h - F_\ell)}{(T_0 + \eta F_\ell)^2} \geq 0, \qquad \frac{\partial S}{\partial T_0} = \frac{\eta(F_\ell - F_h)}{(T_0 + \eta F_\ell)^2} \leq 0.$$

For $F_h > F_\ell$, faster refinement reduces the value of cutting its budget; lower fixed cost increases it. The derivatives hold with non-refinement work fixed. Budget-dependent scan maintenance and resource transitions require the controlled comparisons of sections 7.5 and 7.5.6. Quality and reuse remain constraints (table 7.7 and figure 7.6).

8.2 Two Deep Negative-Result Case Studies

8.2.1 Mailman-style shared sums: arithmetic reuse meets scatter cost

Mailman reuses finite-alphabet matrix–vector sums [20]. We test atomic buckets, private histograms and widths 4/6/8 against packed two-table scoring on frozen Vogue/arXiv/OpenAI-3072 streams, 128 probes, batches 8/1,000. Codes and mathematical scores stay fixed.

The best alternatives reach only 0.187–0.581 of baseline throughput (figure 8.1). Profiles spend most cycles in bucket accumulation: list-local bias repeats destinations; private histograms trade atomics for larger state and shared updates. Narrower groups help some variants, but none wins. This stopped these implementations, not all finite-alphabet reuse.

The pilot emits scores, not neighbours. FP32/FP64 cutoff differences therefore do not establish pipeline equivalence. With scorer ratio s and replaceable fraction f, $1/[(1-f)+f/s]$ is only a replacement model: integration stopped before end-to-end testing. A lossless-dictionary study also stopped at structural statistics. Neither supplies a measured system-frontier point.

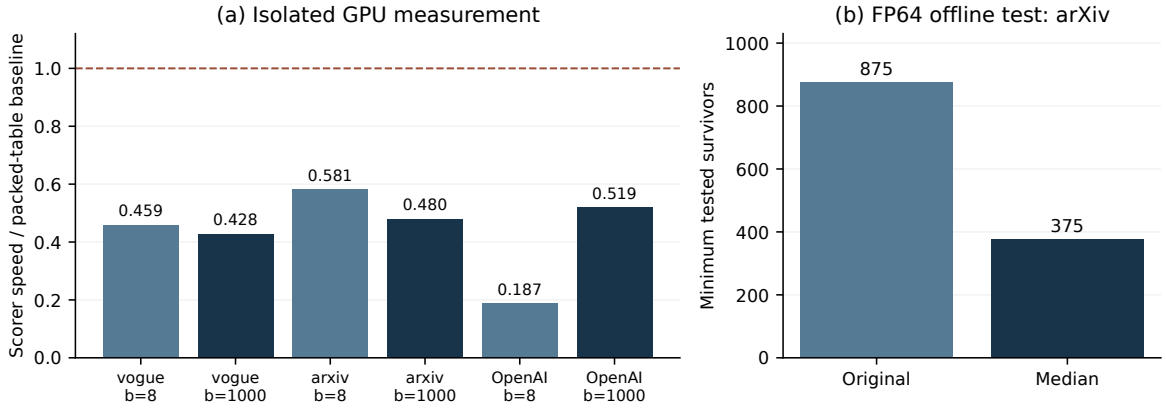


Figure 8.1. Two distinct evidence boundaries. Left: best measured Mailman scorer / packed-table scorer speed on an RTX 5090, with Vogue, arXiv and OpenAI-3072 at two batches and 128 probes; all ratios are below one. Right: minimum survivor counts meeting the original JHQ recall target on arXiv's 400-query FP64 offline test split. The candidate reduction is not a GPU speedup.

8.2.2 Rank-oriented coding: a local gain is not a general replacement

The ranking study replays exported candidates in FP64 with 400/200/400 calibration/validation/test queries. It tests weighted scores, median/list thresholds, learned losses and linear-MMSE decoding. Better reconstruction or inner-product MSE can still worsen nearest-candidate order; changing primary levels also changes residual reference points.

On arXiv test queries, median thresholds cut the minimum tested qualifying survivor count from 875 to 375; the other two workloads fail the same strict criterion. Rotation preserves corpus-mean norm, so Gaussian intuition does not remove bias. Centring may help, but must also preserve the refined-score ceiling. The improvement is workload-specific.

A separate GPU follow-up imports median thresholds/levels and retrains residuals. At arXiv's original $\alpha = 100$ target, gains are about 1.40/1.21 at batches 8/1,000. Vogue stops at recall 0.9630, below target 0.9644; OpenAI-3072 is tied. These three-round results use separate states/queries and are excluded from E1.

8.3 Threats to Internal and External Validity

Table 8.1 consolidates threats to the main interpretations. Chapter 3 gives controls; section 7.8 checks repeats, rebuilding and precision.

Table 8.1. Threats that can reverse or narrow a conclusion.

Claim at risk	Alternative or reversal	Control and remaining test
Execution gain	Changed scores, ties or index state explain an apparent speedup	Strict T4 CPU sorting and table-form diagnostics; finite coverage, with candidate dumps no longer retained
Budget quality	Finite-pool sampling or drift hides damaging queries	Disjoint GT evaluation, full traces, subset-sum checks and FULL enumeration; future-query and arbitrary-subset invariance remain untested
Competitive margin	Grid density, call tails or API overhead changes the ordering	Both recall estimators, retained slow calls and int8 baseline; TF32/FP32 sensitivity; T5 exposes small-batch reversals; cross-device comparisons remain unmeasured
Capacity/build claim	Transient allocations exceed a resident sample	Phase bytes, repeated empty-cache builds and 20 ms device peaks; sub-interval allocations and full retrained frontiers remain unmeasured

Coverage is one GPU, squared-L2 top-10 and six prepared datasets (section 7.2.5). Query removal does not exclude near duplicates; relevance labels are absent. Updates, concurrency, drift, service latency, billion-scale and multi-device capacity remain untested.

8.4 Testable Research Directions

8.4.1 Risk-controlled agreement with a reuse-aware objective

A future policy could minimise calibration-plus-reuse time with valid query-level reference-agreement bounds. It needs matched-regime counts, a fixed reference, explicit sampling/order assumptions and drift-triggered renewal. An absolute GT target additionally needs labels or justified reference-to-GT assumptions.

Compare fixed sampling, reusable rank curves and cost-aware selection on held-out pools, measuring total time and tail agreement. An older four-cell sequential controller lost because extra launches outweighed sample savings. Success must therefore save total time against a quality-matched control while satisfying the stated risk bound.

8.4.2 Capacity-aware residual placement

A residual cache could keep primary codes on-device and fetch missed residual blocks from host memory under a fixed device cap. Hold codes, routing, survivors and arithmetic fixed; vary cap, batch and locality. Measure transfer bytes, hits, overlap, tail latency and throughput. It must extend usable capacity at acceptable cost. This placement study is unimplemented; lowering α does not provide it.

8.5 Conditional Deployment Recommendation

For recall 0.97, E1 supports batch-1,000 throughput; T5 favours CAGRA fp32 on every batch-8 dataset. At batch 32, only OpenAI-3072's near tie changes with the estimator (section 7.2.5). Start with FULL, then validate deployment quality, reuse cost and memory (section 8.3).

9 Conclusion

9.1 Answers to the Four Research Questions

RQ1: Construction and capacity. GPU training, batched encoding, IVF placement and packed arrays produce a searchable resident hierarchy. The largest tested database contains 17.78 million vectors at 20.44 GiB resident device memory; the corresponding supplementary train-plus-add range is 10.54–10.65 s (section 7.7). Its sampled construction peak is 22.58 GiB, while BGE reaches 26.08 GiB at add entry.

RQ2: Correct and efficient execution. The inherited Cartesian code admits exact real-arithmetic table factorisation, and complete-key retention/merge and shared-bound arguments explain the grouped path's selection conditions. Two-group tables lead in 22/24 measured cells. Integrated scheduling achieves 2.727× geometric-mean throughput at batch 8 but only 1.015× at batch 1,000 against the lane-refinement reference (sections 7.3 and 7.6). Thus useful parallelism depends on batch and work distribution. T4's strict CPU reference matches all 55,296 grouped instances in complete-key order; 56 one-block set differences are exclusively cutoff ties (section 7.4.1).

RQ3: Refinement budget and quality. FULL calibration meets the tested relative recall-loss target in 12/12 cells, with maximum observed loss 0.0006. At $S = 128$ reaches 40/64 crossings in one cell (section 7.5). All FULL choices match enumeration, but ten distance-only cells lack a general nesting guarantee. Reference agreement is not population GT recall. Four direct reuse tests show workload-dependent quality/time gains.

RQ4: Competitive operating regions. At recall 0.97 and batch 1,000, JHQ leads on three datasets and is near RaBitQ on arXiv; Vogue needs grouping (section 7.2). OpenAI E1 uses TF32. Cached-model FP32 checks retain advantages over archived D baselines and contemporaneous T1 anchors, without FP32 retraining (section B.4.1). Int8 wins lower targets; RaBitQ wins some higher ones. CAGRA fp32 wins all batch-8 datasets; batch-32 near ties depend on the estimator (section 7.2.5).

9.2 Closing Principle and Outlook

Representation, execution and effort must share a quality target. Next steps are reusable risk-controlled calibration and capacity-aware residual placement (section 8.4).

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A Reproducibility and Experiment Manifest

This appendix specifies the experimental environment and measurement protocol. Available source and measurement materials are listed in section B.5.

A.1 Source, Environment and Trained States

Table A.1 lists run-header/report versions. Environment and JIT settings follow Chapter 3; historical executable and model fingerprints remain incomplete.

Data/GT follow table 3.2. Sweeps reuse trained caches; cold builds do not. Isolated large-baseline reruns replace contaminated builds. Corrected Stella search uses 999 queries; skipped batch-1,000 runs supply no points.

A.2 Measurement Contracts

E1 takes the median of five round QPS values, normally 24 calls each. Warm-up classification reduces calls at 274 points; its first-call conversion/validation interval differs from steady-state timing. Host staging, int8 quantisation and transfers are timed; API/output-memory differences remain. Device-only times are auxiliary.

The 369 repeat-ID flags include tie-order changes; 401 also counts cross-protocol disagreement. Flags alone do not exclude points, and JHQ lacks the same flag. Cold timers exclude constructor/host QR/warm-up. Build sampling includes context at 20 ms; synchronised one-buffer diagnostics are separate.

A.3 Budget Harness Contract

The campaign overrides the fallback grid starting at 100 with one starting at 200. Pools/draws follow Chapter 3. Initial warm-up visits all queries but supplies neither GT nor outputs to agreement; scan calibration uses separate database pseudoqueries.

Fixed arms use one warm-up and three timed calls. Calibration times searches plus $O(Sk^2)$ host comparison, bypassing graph capture; gathering and scan calibration are excluded. p95 uses index $\lfloor 0.95(64-1) \rfloor$. The helper returns an index, so callers must apply `set_alpha(A[index])`. Original selected-arm rates reuse fixed-arm timings.

Original enumeration covers 2,304 draws; supplementary checks add twelve FULL decisions. All choices and 18,432 subset-sum grid counts match. General ordering/invariance conditions remain those of section 6.3 and equation (6.9); GT-risk mapping uses held-out labels.

A.4 Supplementary Evidence

A1 separates first-read, warm-input and synchronised builds. C1 checks four layouts and sampled recoding; FP64 norms use stored rotations, not GPU rotation outputs. A3 checks all OpenAI queries and 10,299 base rows without original entity IDs.

Table A.1. Recorded component provenance. Report and script identities are distinguished from raw run headers.

Component	Recorded source evidence
E1 final pipeline	Environment logs include `a31c22d`; extensions retain separate metadata
Lane refinement (v61)	Raw header `b37efa4`
Grouped scan (v62)	Report `862543a`; run script `e0d0518`
Shared bound (v63)	Source and build flags archived; no inferred run head
Batch scheduling (v65)	Raw header `d425ab8`
Radix probe (v69)	Comparative runs and source archived; binary SHA unrecorded
Budget / table sweep	Raw header `d5e1526`
Large int8 rerun	Stella degree-64 metadata `ad433a4`
Historical budget	Raw header `bf31b41`, v57 family

A2 reassembles extracted sm_80 PTX with CUDA 13.0.88 `ptxas` for sm_120. Aligned eight-byte key accesses support the PTX contract; toolkit SASS is not captured driver-JIT code.

D varies add/query GEMMs on TF32-trained caches against archived baselines. T1 supplies separate same-session anchors (section B.4.1); neither retrains FP32 models.

B reproduces `srand(12345)` decisions. Direct reuse instead uses a separate seed, five rotated-policy rounds and 500 repeated batches. Its timer includes gathering, calibration, budget installation and first-graph setup, but excludes build, scan calibration and warm-up. Schedule startup is reported separately.

A.5 Strict Selection, Small Batches and Artifact Access

T4 v73 differs from v71 by two export blocks. GPU-produced scores feed CPU sorting; bitwise equality covers returned valid primary entries, and padding checks cover valid counts. Table-form checks fix 128 probes, $\alpha = 100$, 256 queries, shared bounds, $g_{\text{scan}} \in \{0, 2\}$ and $g_{\text{LUT}} \in \{1, 2, 4, 8\}$. Candidate dumps were deleted; summaries retain finite counts. Empty/adversarial lists and the unexplained cross-process Vogue discrepancy are separate questions.

T5 uses fixed small synchronous batches and five host-to-host rounds, including null-call costs and int8 query quantisation. JHQ/CAGRA reuse E1 grids; RaBitQ has a separate sweep. Both threshold estimators retain qualifying endpoints. APIs/warm-ups differ; service concurrency is untested.

GPU reproduction requires original vectors, matching trained states and builds, and the specified environment. Section B.5 identifies which materials are available.

B Supplementary Results and Correctness Checks

Parameter coverage and secondary checks follow below; protocol is in Appendix A.

B.1 Parameter Coverage

The E1 comparison contains 1,590 valid measured configurations, with coverage in table B.1. The sets below describe observed parameter values; their Cartesian products are not assumed to have been measured.

Table B.1. Exact E1 configuration counts, including coverage extensions. Counts are parameter points, not independent index builds.

Family	Vogue	arXiv	OpenAI-1536	OpenAI-3072	Total
JHQ-GPU v69	27	29	37	39	132
CAGRA fp32	64	64	64	64	256
CAGRA int8	64	64	64	64	256
IVF-RaBitQ	132	132	264	198	726
IVF-PQ	54	54	54	54	216
GPU brute force	1	1	1	1	4
Total	342	344	484	420	1590

JHQ-GPU. All E1 points use $B = B_r = 8$, $M = d/8$, $k = 10$ and batch 1,000. Vogue uses 4,096 IVF lists; the other three use 8,192. Vogue observes probes 16/32/64/96/128/192/256/384/512 and $\alpha \in \{16, 32, 64, 128\}$ over 27 tuples. ArXiv adds probe 1,024 and uses $\alpha \in \{25, 50, 100, 200\}$ over 29 tuples. Both OpenAI grids use probes 32/64/128/256/512/1,024; their observed multipliers are respectively $\{4, 8, 16, 32, 64, 128, 256, 512\}$ and $\{2, 4, 8, 16, 32, 64, 128, 256\}$, over 37 and 39 tuples. The reported scheduled path uses four groups on Vogue and one block elsewhere. OpenAI uses TF32 rotation/centroid GEMMs in historical v69; the v72 FP32 sensitivity is a separate family (section 7.2.3).

CAGRA fp32 and int8. Each representation has 64 points per dataset: graph degrees 32/64, intermediate degree 128, `ivf_pq` construction, `itopk_size` in 32/64/96/128/192/256/384/512 and search width 1/2/4/8. Int8 records quantile 0.99 for dataset scaling and includes query quantisation in the host search boundary.

IVF-PQ. Eight-bit subcodes use 192/384/768-byte codes at dimension 768, and 384/768/1,536-byte codes on OpenAI. Each length has fp16/fp32 lookup tables and probes 16/32/64/96/128/192/256/384/512: 54 points per dataset. The recorded refinement ratio is one and list counts match JHQ. Code lengths and memory budgets differ.

IVF-RaBitQ and brute force. RaBitQ sweeps 4/8/9 bits and probes 16/32/64/96/128/192/256/384/512/768/1,024. Modes are LUT16/LUT32/QUANT4/QUANT8 except OpenAI-3072 omits LUT16. OpenAI adds 4,096-list indexes to default 8,192-list coverage; exact combinations remain in the registry. Each dataset has one FP32 squared-L2 brute-force point. Missing coverage is not failure.

Auxiliary grids. LUT sweeps six datasets, four probe counts (8/32/128/512) and $g_{\text{LUT}} \in \{1, 2, 4, 8\}$ at $\alpha = 100$: 24 cells, 96 arms. Budget evaluation uses probes 128/512 and $A = (200, 100, 64, 32, 16, 8, 4, 2)$ on the same six datasets, with 64 draws at each $S \in \{32, 64, 128\}$ and $\epsilon = 1$. FULL separately uses all 500 calibration queries. Held-out counts are 500, or 499 for Stella. Boundaries are specified in sections A.3 and A.4.

B.2 Correctness and Numerical Evidence

Table B.2 separates numerical, algorithmic and memory evidence. Chapter 5's ordering arguments concern computed scores under their stated contracts; raw-vector recall is evaluated separately.

Table B.2. Validation coverage and the object actually checked.

Boundary	Completed evidence and scope
Representation	Four all-row offset/ID audits; sampled raw-input FP64 recoding and stored-matrix geometry; TF32 deviations identified
Cartesian LUTs	Real-arithmetic factorisation; 96 timed arms; T4 per-candidate FP32 error and ranking diagnostics across four table forms
Global selection	T4: 270 configurations, 69,120 query instances; 55,296 grouped instances match strict CPU complete-key sorting in order
Shared bound	48 paired checks; extracted scalar 64-bit PTX and toolkit SASS support whole-key access; actual driver SASS not captured
Scheduling	250 variant/configuration and 234,320 query comparisons, retaining distinctions between ID sets and tie order
Probe selector	1,792 synthetic cases with zero failures; real-query ID/offset checks, relative to computed routing scores
Budget search	2,304 monotone sampled curves and 12 FULL enumeration matches; 18,432 sampled grid counts reproduce full-pool subset sums

The smaller representation-audit databases have identical primary and residual bytes across one-/two-pass builds when joined by ID. Corrections differ by at most 8.94×10^{-8} and 1/1/40 rows change list assignment; Stella validates its natural two-pass path (section 4.6.2). T4 supplies the strict computed-score selection reference (section A.5). The controlled same-model Vogue cross-process diagnosis remains unperformed: archived processes with matching probe/primary dumps still report final recall 0.9642/0.9644. Within-run selection checks do not localise that separate anomaly.

B.3 Secondary Results and Evidence Status

Adverse regimes. Worst scheduling regret is 1.217 at batch 256; Vogue E1 ratios span 0.895–1.175 (section 7.6). T5 CAGRA fp32 wins every batch-8 dataset at recall 0.97; batch-32 ties depend on the estimator (section 7.2.5).

B.4 Construction Capacities and Precision Anchors

Table B.3 separates staging from assignment scratch. Defaults use three buffers, 65,536 add rows and 128 MiB per input/rotation array; these are capacities, not current batch sizes or process peaks.

Table B.3. Default construction-buffer capacities. The last column is $4n_b n_{\text{list}} a_{\text{assign}}/2^{30}$; substituting a_{in} would give an incorrect general model.

d	n_{list}	a_{in}	a_{assign}	Dots GiB
768	4,096	43,690	8,192	0.375
1,024	32,768	32,768	32,768	12.000
1,536	8,192	21,845	32,768	3.000
3,072	8,192	10,922	32,768	3.000

B.4.1 Contemporaneous precision anchors

T1 remeasures JHQ and baselines near recall 0.97 in one session with E1 pools. Twelve configuration pairs per OpenAI dataset bracket baselines with two five-round JHQ repetitions; QPS combines their medians geometrically. FP32/TF32 means are 0.956/0.964. Both reuse TF32-trained caches and change add/query arithmetic.

Table B.4. T1 at recall 0.97 and batch 1,000. Baseline columns report interpolation / strict measured threshold; the FP32/TF32 column is interpolated. All methods here are contemporaneous; these values are separate from D's cross-session baseline ratios.

Dataset	FP32/TF32 at 0.97	FP32 / RaBitQ	FP32 / CAGRA
OpenAI-1536	0.980	1.126 / 1.053	1.752 / 1.633
OpenAI-3072	0.973	1.328 / 1.316	1.825 / 2.019

Both estimators retain numerical FP32 advantages. OpenAI-3072's RaBitQ bracket (0.9669–0.9796 recall) exposes grid sensitivity. These anchors neither replace E1 nor retrain FP32 models.

B.5 Artifact Access

The project artifact archive provides source/measurement snapshots and selected T4/T5, T1 and A2 inputs. Complete vectors, model states and historical executables are absent; GPU reproduction needs those plus the matching environment.